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United States Patent	12399779
Kind Code	B2
Date of Patent	August 26, 2025
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Devices, methods, and computer-readable media to back-up parity information

Abstract

Embodiments described herein provide a data storage device that backs-up parity information. The data storage device includes a memory and a data storage device controller coupled to the memory and including a storage sub-system and a security sub-system. The storage sub-system includes parity information within a second memory. The data storage device further includes a bus for transferring data between the data storage device controller and an external device in communication with the data storage device controller. The data storage device controller is configured to initiate a parity information back-up request, send, from the second memory of the storage sub-system, the parity information to the security sub-system based on the initiation of the parity information back-up request, and control the security sub-system to transmit the parity information to the external device.

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Appl. No.: 18/448116

Filed: August 10, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240248799 A1	Jul. 25, 2024

Related U.S. Application Data

us-provisional-application US 63481315 20230124

Publication Classification

Int. Cl.: G06F11/00 (20060101); G06F11/10 (20060101); G06F11/14 (20060101)

U.S. Cl.:

CPC G06F11/108 (20130101); G06F11/1456 (20130101);

Field of Classification Search

CPC: G06F (11/108); G06F (11/1456); G06F (3/0619); G06F (3/055); G06F (3/0679)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application claims priority to and the benefit of U.S. Provisional Patent Application No. 63/481,315, filed on Jan. 24, 2023, the entire contents of which is incorporated herein by reference.

FIELD

(1) This application relates generally to data storage devices, and more particularly, to devices, methods, and computer-readable media for backing up a parity information generated by the data storage devices.

BACKGROUND

(2) Flash memory is commonly used in everyday devices including smart telephones, computers, disk drives, and other electronic devices. Compared to other types of memory, flash memory is

smaller in size, has relatively low power consumption, and is easier to interface electronically. However, flash memory may have failures, for example, program failures, silent read failures, and the like. These failures may result in data loss that may reduce the reliability of the data storage device.

SUMMARY

(3) Conventional designs implement parity information (e.g., XOR snapshots) as a Flash Translation Layer (FTL) data in a static random access memory (SRAM) of a data storage device controller. During the run-time of a data storage device controller, the FTL transfers the parity information to NAND, thus consuming device time and dropping device performance. For example, bottlenecks may be caused by the FTL and a low layer (LL) transferring the parity information to the NAND and writing the parity information.

(4) Additionally, conventional designs are prone to loss of data due to the parity information being stored in the SRAM, which is volatile and at risk of losing data. For example, the parity information may be corrupted based on the quality-grade and characteristics of the SRAM used in the data storage device controller and, thus, the parity information may be lost. As a result, if the parity information is required, conventional designs include the FTL performing a read operation from flash memory which provides additional strain to the data storage device.

(5) The parity information may be stored in one or more SRAM banks of the SRAM, which is referred to herein as “XRAM” and is an abbreviation of XOR and SRAM. However, the storage of the parity information is not limited to XRAM and may be stored elsewhere in other types of memories (e.g., DRAM) that may be internal or external to the data storage device controller. Accordingly, there is a need to protect host data from the failures and issues described above.

(6) To solve these and other problems, the embodiments described herein provide a stored back-up of parity information in a secured cached database using a Bluetooth® interface and a security processor. In particular, the embodiments described herein provide a data storage device controller with a user authentication-based access (for example, via Bluetooth®) for storing a back-up of the parity information in a user application. The data storage device controller includes a Storage Sub-System (StorSS) and a Security Sub-System (SecSS) that controls a Bluetooth® transceiver and provides the user authentication. Additionally, the embodiments described herein provide the technical advantage of increased device quality of service.

(7) One embodiment of the present disclosure includes a data storage device. The data storage device includes a memory and a data storage device controller coupled to the memory and including a storage sub-system and a security sub-system. The storage sub-system includes parity information within a second memory. The data storage device further includes a bus for transferring data between the data storage device controller and an external device in communication with the data storage device controller. The data storage device controller is configured to initiate a parity information back-up request, send, from the second memory of the storage sub-system, the parity information to the security sub-system based on the initiation of the parity information back-up request, and control the security sub-system to transmit the parity information to the external device.

(8) Another embodiment of the present disclosure includes a method of backing-up parity information. The method includes initiating, with an electronic processor, a parity information back-up request, controlling, with the electronic processor, a storage sub-system coupled to the electronic processor to send parity information from a memory of the storage sub-system to a security sub-system coupled to the electronic processor based on the initiation of a parity information back-up request, and controlling, with the electronic processor, the security sub-system to transmit the parity information to an external device.

(9) Yet another embodiment of the present disclosure includes a non-transitory computer readable medium coupled to an electronic processor. The non-transitory computer readable medium includes computer executable instructions stored in therein that, when executed by the electronic processor,

enable the electronic processor to initiate a parity information back-up request, control a storage sub-system coupled to send parity information from a memory of a storage sub-system to a security sub-system coupled to the electronic processor based on the initiation of a parity information back-up request, and control the security sub-system to transmit the parity information to an external device.

(10) Various aspects of the present disclosure provide for improvements data storage devices. The present disclosure can be embodied in various forms, including hardware or circuits controlled by software, firmware, or a combination thereof. The foregoing summary is intended solely to give a general idea of various aspects of the present disclosure and does not limit the scope of the present disclosure in any way.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is block diagram of a system including a data storage device, a host device, and an external device, in accordance with some embodiments of the disclosure.

(2) FIG. 2 is a diagram illustrating a first example back-up system, in accordance with some embodiments of the disclosure.

(3) FIG. 3 is a diagram illustrating a second example back-up system, in accordance with some embodiments of the disclosure.

(4) FIG. 4 is a block diagram of the external device of FIG. 1, in accordance with some embodiments of the disclosure.

(5) FIG. 5 is a flowchart illustrating a process for outputting parity information to an external device, in accordance with some embodiments of the disclosure.

(6) FIG. 6 is a flowchart illustrating a method for storing parity information in the security sub-system memory, in accordance with some embodiments of the disclosure.

(7) FIG. 7 is a flowchart illustrating a process for retrieving parity information, in accordance with some embodiments of the disclosure.

(8) FIG. 8 is a flowchart illustrating a process for storing parity information in the memory of an external device, in accordance with some embodiments of the disclosure.

(9) FIG. 9 is a flowchart illustrating a process for outputting parity information to the security sub-system, in accordance with some embodiments of the disclosure.

DETAILED DESCRIPTION

(10) In the following description, numerous details are set forth, such as data storage device configurations, controller operations, and the like, in order to provide an understanding of one or more aspects of the present disclosure. It will be readily apparent to one skilled in the art that these specific details are merely exemplary and not intended to limit the scope of this application. In particular, the functions associated with the data storage device controller can be performed by hardware (for example, analog or digital circuits), a combination of hardware and software (for example, program code or firmware stored in a non-transitory computer-readable medium that is executed by a processor or control circuitry), or any other suitable means. The following description is intended solely to give a general idea of various aspects of the present disclosure and does not limit the scope of the disclosure in any way. Furthermore, it will be apparent to those of skill in the art that, although the present disclosure refers to NAND flash, the concepts discussed herein are applicable to other types of solid-state memory, such as NOR, PCM (“Phase Change Memory”), ReRAM, MRAM, etc.

(11) FIG. 1 is block diagram of a system including a data storage device, a host device, and an external device, in accordance with some embodiments of the disclosure. In the example of FIG. 1, the system **100** includes a data storage device **102**, a host device **120**, and an external device **130**.

In some embodiments, the system **100** may include multiple data storage device and/or multiple external devices. The data storage device **102** includes a controller **106** (referred to hereinafter as “data storage device controller”) and a memory device **104** (e.g., non-volatile memory) that is coupled to the data storage device controller **106**.

(12) One example of the structural and functional features provided by the data storage device controller **106** is illustrated in FIG. **1** in a simplified form. One skilled in the art would also recognize that the data storage device controller **106** may include additional modules or components other than those specifically illustrated in FIG. **1**. Additionally, although the data storage device **102** is illustrated in FIG. **1** as including the data storage device controller **106**, in other implementations, the data storage device controller **106** is instead located separate from the data storage device **102**. As a result, operations that would normally be performed by the data storage device controller **106** described herein may be performed by another device that connects to the data storage device **102**.

(13) The data storage device **102** and the host device **120** may be operationally coupled via a connection (e.g., a communication path), such as a bus or a wireless connection. In some examples, the data storage device **102** may be embedded within the host device **120**. Alternatively, in other examples, the data storage device **102** may be removable from the host device **120** (i.e., “removably” coupled to the host device **120**). As an example, the data storage device **102** may be removably coupled to the host device **120** in accordance with a removable universal serial bus (USB) configuration. In some implementations, the data storage device **102** may include or correspond to a solid-state drive (SSD), which may be used as an embedded storage drive (e.g., a mobile embedded storage drive), an enterprise storage drive (ESD), a client storage device, or a cloud storage drive, or other suitable storage drives.

(14) The data storage device **102** and the external device **130** may be operationally coupled via a connection (e.g., a communication path), such as a bus or a wireless connection. In some examples, the data storage device **102** communicates with the external device **130** via a Bluetooth® connection. In some embodiments, parity information may be sent between the data storage device **102** and the external device **130**, as will be described in greater detail below.

(15) The data storage device **102** may be configured to be coupled to the host device **120** via the communication path, such as a wired communication path and/or a wireless communication path. For example, the data storage device **102** may include an interface **116** (e.g., a host interface) that enables communication via the communication path between the data storage device **102** and the host device **120**, such as when the interface **116** is communicatively coupled to the host device **120**. In some embodiments, wireless communication may be conducted using Bluetooth®. For example, the data storage device controller **106** may include a Bluetooth® communication component (e.g., Bluetooth® controller **115**) that communicates with a Bluetooth® communication component of the host device **120**. Parity information may be sent between the data storage device **102** and the host device **120**.

(16) The host device **120** may include an electronic processor and a memory. The memory may be configured to store data and/or instructions that may be executable by the electronic processor. The memory may be a single memory or may include one or more memories, such as one or more non-volatile memories, one or more volatile memories, or a combination thereof. In some embodiments, the memory of the host device **120** may include a database for storing parity information from the data storage device **102**. The host device **120** may issue one or more commands to the data storage device **102**, such as one or more requests to erase data at, read data from, or write data to the memory **104** of the data storage device **102**. Additionally, the host device **120** may issue one or more vendor specific commands to the data storage device **102** to notify and/or configure the data storage device **102**. For example, the host device **120** may be configured to provide data, such as user data, to be stored at the memory **104** or to request data to be read from the memory **104**. The host device **120** may include a mobile smartphone, a music player, a video player, a gaming

console, an electronic book reader, a personal digital assistant (PDA), a computer, such as a laptop computer or notebook computer, any combination thereof, or other suitable electronic device.

(17) The host device **120** communicates via a memory interface that enables reading from the memory device **104** and writing to the memory device **104**. In some examples, the host device **120** may operate in compliance with an industry specification, a Secure Digital (SD) Host Controller specification, or other suitable industry specification. The host device **120** may also communicate with the memory device **104** in accordance with any other suitable communication protocol (e.g., Bluetooth®). For example, the host device **120** may receive parity information via Bluetooth® communication.

(18) The external device **130** may include an application **122** and an electronic processor, such as electronic processor **400** (FIG. 4). The application **122** may include a database **124** that is a secured cached memory. In some embodiments, the application **122** hosts the database **124**. The internal mapping of the database **124** and the contents of the database **124** may be taken care by the application **122** and/or the database **124**, internally. The data storage device **102** may send and retrieve parity information from the external device **130** to back-up the parity information in the external device **130**. For example, parity information may include a XOR snapshot, such as XOR snapshot **112**. In some embodiments, the XOR snapshot **112** may be parity information that corresponds to data. In some embodiments, the data storage device **102** may send and retrieve any device data from the external device **130** to back-up the device data in the external device **130**. The external device **130** may execute the application **122** which may store the XOR snapshot **112** in the database **124** upon receipt of the XOR snapshot **112** from the security sub-system **114**. In some embodiments, the application **122** is stored in a memory, such as memory **404** (FIG. 4) of the external device **130**. The external device **130** may include a mobile smartphone, a music player, a video player, a gaming console, an electronic book reader, a personal digital assistant (PDA), a computer, such as a laptop computer or notebook computer, a dedicated parity information storage device, any combination thereof, or other suitable electronic device. For example, the dedicated parity information storage device may be a wireless, external device that is dedicated to receiving and storing the parity information. The external device **130** may include a security setting that may lock and unlock the database **124** based on an input. For example, the input may be one of a fingerprint reading, a facial recognition reading, or a passcode that is input. In some embodiments, an input may be received before execution of the application **122**. The dedicated parity information storage device may be an addition onto the system **100**. In some embodiments, the application **122** is pre-installed on the dedicated parity information storage device. Alternatively, in some embodiments, the application **122** is installed on the dedicated parity information storage device by a user. For example, the application may be installed via a user interface of the dedicated parity information storage device, or via an external device, such as host device **120** or external device **130**.

(19) The memory device **104** of the data storage device **102** may include a non-volatile memory (e.g., NAND, NOR, 3D NAND family of memories, or other suitable memory). In some examples, the memory device **104** may be any type of flash memory. For example, the memory device **104** may be two-dimensional (2D) memory or three-dimensional (3D) flash memory. The memory device **104** may include one or more memory dies **105**. Each of the one or more memory dies **105** may include one or more blocks (e.g., one or more erase blocks). Each block may include one or more groups of storage elements, such as a representative group of storage elements **107A-107N**. The group of storage elements **107A-107N** may be configured as a word line. The group of storage elements **107** may include multiple storage elements, such as a representative storage elements **109A** and **109N**, respectively.

(20) The memory device **104** may include support circuitry, such as read/write circuitry **111**, to support operation of the one or more memory dies **105**. Although depicted as a single component, the read/write circuitry **111** may be divided into separate components of the memory device **104**,

such as read circuitry and write circuitry. The read/write circuitry **111** may be external to the one or more memory dies **105** of the memory device **104**. Alternatively, one or more individual memory dies may include corresponding read/write circuitry that is operable to read from and/or write to storage elements within the individual memory die independent of any other read and/or write operations at any of the other memory dies.

(21) The data storage device **102** includes the data storage device controller **106** coupled to the memory device **104** (e.g., the one or more memory dies **103**) via a bus, an interface (e.g., interface circuitry), another structure, or a combination thereof. For example, the bus may include multiple distinct channels to enable the data storage device controller **106** to communicate with each of the one or more memory dies **103** in parallel with, and independently of, communication with the other memory dies **103**. In some implementations, the memory device **104** may be a flash memory.

(22) The data storage device controller **106** is configured to receive data and instructions from the host device **120** and/or the external device **130** and to send data to the host device **120** and/or the external device **130**. For example, the data storage device controller **106** may send data to the host device **120** via the host interface **116**, and the data storage device controller **106** may receive data from the host device **120** via the host interface **116**. While data is being sent to and from the host device **120** via the host interface, parity information (e.g., the XOR Snapshot **112**) may be sent to and from the data storage device controller **106** to the external device **130** via a wireless connection (e.g., Bluetooth®). The data storage device controller **106** is configured to send data and commands to the memory device **104** and to receive data from the memory device **104**. For example, the data storage device controller **106** is configured to send data and a write command to cause the memory device **104** to store data to a specified address of the memory device **104** via the memory interface **118**. The write command may specify a physical address of a portion of the memory device **104** (e.g., a physical address of a word line of the memory device **104**) that is to store the data, as well as a size (for example, 2 k, 4 k, etc.) of the data to be written.

(23) The data storage device controller **106** is configured to send a read command to the memory device **104** to access data from a specified address of the memory device **104**. The read command may specify the physical address of a region of the memory device **104** (e.g., a physical address of a word line of the memory device **104**). The data storage device controller **106** may also be configured to send data and commands to the memory device **104** associated with background scanning operations, garbage collection operations, and/or wear-leveling operations, or other suitable memory operations.

(24) The data storage device controller **106** may include a storage sub-system **108**, a security sub-system **114**, and an electronic processor **119**. In some embodiments, the storage sub-system **108** and the security sub-system **114** may include their own, dedicated electronic processors.

(25) The storage sub-system **108** may include a storage memory **110** (for example, SRAM including the XRAM, a read-only memory (“ROM”), a non-transitory computer readable medium, or a combination thereof). In some embodiments, the storage memory **110** may include the XRAM storing parity information (e.g., an XOR snapshot **112**) that may be used in a data recovery process. Data recovery in the storage memory **110** is possible through a XOR computation process executed by the storage sub-system **108**. For example, the XOR computation process includes finding parity data (P) for data, dataB, dataC according to the following: $P = (data \oplus ()) \oplus dataB \oplus dataC$. To retrieve dataB from the storage memory **110**, the storage subsystem **108** may use the following: $dataB = (P \oplus ()) \oplus data \oplus dataC$. In some embodiments, access to the XRAM is contingent on traditional ASIC memory usage guidelines. The storage memory **110** may store the XOR Snapshot **112** as described in greater detail with respect to FIGS. 2 and 3.

(26) The security sub-system **114** may include a Bluetooth® controller **115** and a memory **117**. The Bluetooth® controller **115** may include a transceiver for communicating with the external device **130** via the Bluetooth® protocol. In some embodiments, the security sub-system includes a Wi-Fi

transceiver for communicating with the external device **130**. The memory **117** may be an SRAM including the XRAM. In some embodiments, the parity information (e.g., the XOR snapshot **112**) may be backed-up by the security sub-system **114** in the memory **117**. The storage sub-system **108** and the security sub-system **114** may communicate with each other via a wired or wireless connection. For example, the storage sub-system **108** and the security sub-system **114** may communicate via a bus within the data storage device controller **106**. The electronic processor **119** may facilitate communication between the storage sub-system **108** and the security sub-system **114**. (27) The electronic processor **119** of the data storage device controller **106** may send a memory operation (e.g., a write command) to the memory device **104** to cause the read/write circuitry **111** to write the XOR snapshot **112** to a storage element of the memory device **104**. For example, the data storage device controller **106** may send the write command to the memory device **104** in response to receiving a request for write access from the host device **120** and/or the external device **130**. Additionally, the electronic processor **119** may send a memory operation (e.g., a store command and/or a retrieve command) from the Bluetooth® controller **115** to the storage memory **110**. The electronic processor **119** may also send a memory operation (e.g., a retrieve command and/or a store command) from the Bluetooth® controller **115** to the external device **130**. In the case that the storage sub-system **108** and the security sub-system **114** include their own electronic processors, they may be substantially the same as electronic processor **119**.

(28) As noted above, embodiments described herein provide parity information back-up, such as a XOR snapshot **112** back-up using a Bluetooth® transceiver, such as the transceiver of Bluetooth® controller **115**. The XOR snapshot **112** back-up described herein will be described with reference to FIGS. 2 and 3.

(29) FIG. 2 illustrates a first example back-up system **200** for backing up parity information, according to some embodiments. The system **200** includes the memory device **104**, the storage sub-system **108**, the security sub-system **114**, and the application **122**. In some embodiments, the system **200** may not include the application **122** and may instead back up the parity information in the security sub-system **114**. The system backs up the parity information (e.g., the XOR snapshot **112**) using the Bluetooth® controller **115** of the security sub-system **114**, which mimics the write operation to the memory device **104**. In some embodiments, the XOR snapshot **112** is a part of the storage memory **110** of the storage sub-system **108**. In some embodiments, the XOR snapshot **112** is backed up from the storage memory **110** of the storage sub-system **108** to the database **124** of the application **122** via the Bluetooth® controller **115** when the application **122** is executed by one of the host device **120** and the external device **130**. For example, the XOR snapshot **112** may be backed up at a fixed time interval and/or during a power down of the data storage device **102**. In some embodiments, the application **122** may be stored in a memory of the host device **120**. Alternatively, or additionally, the application **122** may be stored in a memory (e.g., memory **404**) of the external device **130**.

(30) The storage sub-system **108** may include a FTL that is executed by the storage sub-system **108** to generate the parity information (e.g., the XOR snapshot **112**) based on requirements (e.g., system performance and system power). In some embodiments, the parity information is generated through a low-density parity-check code (LDPC) engine as part of an encode operation (e.g., executed by the storage sub-system **108**). In some embodiments, the generated parity information may include complete logical block addressing (LBA) range and internal device data (e.g., customer data stored in the data storage device **102**). Alternatively, in some embodiments, internal device data may be the generated parity information. In some embodiments, the XOR snapshot is in the range of 32-64 kilobits (KB). The computed XOR snapshot **112** is then stored in the storage memory **110**. Conventionally, the FTL writes the XOR snapshot **112** to the memory device **104** to back-up the XOR snapshot **112**. For example, the FTL of the storage sub-system **108** writes the XOR snapshot **112** to NAND of the memory device **104**. In some embodiments, an error correction code (ECC) engine accumulates the data to be the XOR snapshot **112** when requested by the FTL. However,

writing the XOR snapshot **112** to the NAND of the memory device **104** may stall or delay other data storage device **102** operations to accommodate the XOR snapshot **112** due to the data path transfers of the NAND needing to be in a synchronized state to avoid race conditions and incorrect write to the storage memory **110** from the NAND of the memory device **104**. In order to mitigate the stalling of the data storage device **102**, the XOR snapshot **112** may be backed up to the database **124** using Bluetooth® via the security sub-system **114**.

(31) When the XOR snapshot **112** is backed up (e.g., at a fixed time interval and/or during a power down), the storage sub-system **108** initiates a parity information request to the security sub-system **114**. The security sub-system **114** may receive the XOR snapshot **112** from the storage sub-system **108** and then initiate a back-up in the database **124** of the application **122** via Bluetooth®. For example, the Bluetooth® controller **115** may store the XOR snapshot **112** in the database **124** when the application **122** is executed by one of the host device **120** and the external device **130** (FIG. 3). In some embodiments, the security sub-system **114** may add security measures when storing the XOR snapshot to the database **124** and when retrieving the XOR snapshot **112** from the database **124**. For example, the security sub-system **114** may encrypt the XOR snapshot **112** using cryptography protocols such as RSA®, elliptic curve cryptography, Advanced Encryption Standard-XTS (AES-XTS), AES-CBC, or other suitable encryption algorithms. Backing up the XOR snapshot **112** in the database **124** provides a dedicated memory for the XOR snapshot **112** without relying on the NAND of the memory **104**, thus, mitigating any bottlenecking with respect to other memory operations.

(32) In some embodiments, the security sub-system **114** may include a memory, such as memory **117** (FIG. 1), that the XOR snapshot **112** is stored in, in addition to, in place of, or while waiting to back up the XOR snapshot **112** to the database **124**. Backing up the XOR snapshot **112** to the database **124** and the memory **117** of the security sub-system **114** may reduce latency with respect to the NAND of the memory **104**.

(33) FIG. 3 illustrates a second example back-up system **300** for backing up parity information, according to some embodiments. The system **300** includes the memory device **104**, the storage sub-system **108**, the security sub-system **114**, and the external device **130**. The external device **130** will be described in detail with respect to FIG. 4, below. The system backs up parity information (e.g., the XOR snapshot **112**) to the external device **130**, using the Bluetooth® controller **115** of the security sub-system **114**, which mimics the internal write operation to the memory device **104** of the data storage device **102**. In some embodiments, the XOR snapshot **112** is a part of the storage memory **110** of the storage sub-system **108**. The XOR snapshot **112** is backed up from the storage memory **110** of the storage sub-system **108** to the database **124** of the application **122** via the Bluetooth® controller **115** in response to a request from the storage sub-system **108** to the security sub-system **114**. In some embodiments, the XOR snapshot **112** may be backed up at a fixed time interval and/or during a power down of the data storage device **102**. For example, the fixed time interval may be in the range of 0.5 seconds (“s”) and 30 s.

(34) When the XOR snapshot **112** is backed up (e.g., at a fixed time interval and/or during a power down), the storage sub-system **108** initiates a parity information back-up request to the security sub-system **114**. The security sub-system **114** may obtain the XOR snapshot **112** and then initiate a back-up in the external device **130** via Bluetooth®. For example, the Bluetooth® controller **115** may output the XOR snapshot **112** to the external device **130**, which executes the application **122**, thus, storing the XOR snapshot **112** in the database **124**. In some embodiments, the security sub-system **114** may add security measures when storing the XOR snapshot to the external device **130** and when retrieving the XOR snapshot **112** from the external device **130**. For example, the security sub-system **114** may encrypt the XOR snapshot **112**. Backing up the XOR snapshot **112** in the external device **130**, and in particular the database **124**, provides a dedicated memory for the XOR snapshot **112**, thus, mitigating any bottlenecking from other memory operations.

(35) FIG. 4 is a block diagram of the external device **130**, according to some embodiments. The

external device **130** may be a separate device from the data storage device **102**. For example, the external device **130** may be an external hard drive, a server that communicates with the data storage device **102** via Bluetooth®, a mobile phone, a computer or any other device with wireless communication capabilities. In some embodiments, the external device **130** may receive parity information from multiple data storage devices. For example, one external device **130** may be used to store a first XOR snapshot from a first data storage device and a second XOR snapshot from a second data storage device. The external device **130** may include an electronic processor **400**, a wireless communication controller **402**, a memory **404**, and an input/output (I/O) interface **406**. The memory **404** may store the application **122**, which may be executed by the electronic processor **400** to facilitate storing of the parity information received from Bluetooth® controller **115** of the security sub-system **114** in the memory **404**. In some embodiments, the application **122** may include the database **124** for storing the XOR snapshot **112** received from the Bluetooth® controller **115**. In some embodiments, the application **122** may be downloadable to the external device **130**. Alternatively, in some embodiments, the application **122** may be pre-downloaded to the memory **404** of the external device **130**. In some embodiments, the memory **404** may store the first XOR snapshot from the first data storage device and the second XOR snapshot from the second data storage device. For example, the database **124** may store both the first XOR snapshot and the second XOR snapshot. In addition to storing the XOR snapshot **112**, the memory **404** may be configured to store data and/or instructions that may be executable by the electronic processor. For example, the memory **404** may be a random access memory (“RAM”). The application **122** may store the XOR snapshot **112** in the database **134** using Bluetooth® protocol. The memory may be a single memory or may include one or more memories, such as one or more non-volatile memories, one or more volatile memories, or a combination thereof. In some embodiments, the external device **130** may include any number of components and is not limited to the components illustrated.

(36) The electronic processor **400** may facilitate communication between the Bluetooth® controller **115** of the security sub-system **114** and the wireless communication controller **402**. The wireless communication controller **402** enables the external device **130** to communicate with the data storage device **102** that includes the Bluetooth® controller **115** of the security sub-system **114**. In some embodiments, the wireless communication controller **402** stores the application **122** and database **124**. In some embodiments, the wireless communication controller **402** is a Bluetooth® controller. The wireless communication controller **402** communicates with the security sub-system **114** employing the Bluetooth® protocol. Therefore, in the illustrated embodiment, the external device **130** and the data storage device **102** are within a communication range (i.e., in proximity) of each other while they exchange data. In some embodiments, the wireless communication controller **402** communicates with the wireless transceiver of the security sub-system **114** over a Wi-Fi network.

(37) The I/O interface **406** may receive input from a graphical user interface. In some embodiments, the graphical user interface may display a first screen of the application **122**. For example, a user may set a fixed time interval for backing up the parity information (e.g., XOR snapshot **112**) via the application displayed on the graphical user interface. The I/O interface **406** may receive input and the electronic processor **400** may facilitate the back-up of the parity information received by the wireless communication controller **402** from the Bluetooth® controller **115**, in the database **124**. In some embodiments, the I/O interface **406** can simultaneously receive inputs as the parity information is provided to the database **124** via Bluetooth®.

(38) FIG. 5 is a flowchart illustrating a method **500** for outputting parity information to an external device, according to some embodiments. It is understood that the process **500** may be stored in a memory, such as memory device **104**, and executed by a processor, such as processor **119**. However, it is contemplated that some or all of the process **500** may be performed on the memory device **104**, such as via the read/write circuitry **111**. Accordingly, while the process **500** is described

in regards to the controller **106** and in particular, the security sub-system **114** described above, it is contemplated that the process **500** may be at least partially performed by the memory device **104** or other suitable memory component.

(39) The security sub-system **114** receives parity information from the storage sub-system **108** (at block **502**). For example, the parity information may be a XOR snapshot, such as XOR snapshot **112**. In some embodiments, the security sub-system **114** receives parity information from the storage sub-system **108** in response to the storage sub-system **108** receiving a parity information back-up request. In some embodiments, the XOR snapshot **112** is received by the Bluetooth® controller **115** of the security sub-system **114**. For example, the memory **110** of the storage sub-system **108** may send the XOR snapshot to the Bluetooth® controller **115** and, in particular, a Bluetooth® transceiver of the Bluetooth® controller **115**.

(40) Once the security sub-system **114** receives the parity information from the storage sub-system **108**, the security sub-system **114** outputs the parity information to an external device, such as external device **130** (at block **504**). In some embodiments, the security sub-system **114** provides the XOR snapshot **112** to the external device **130** via the Bluetooth® transceiver of the Bluetooth® controller **115**. Alternatively, or additionally, in some embodiments, the security sub-system **114** provides the XOR snapshot **112** to the host device **120**. In some embodiments, the external device **130** executes an application (e.g., application **122**) that receives the XOR snapshot **112**. For example, when the external device **130** executes the application **122**, the XOR snapshot **112** is stored in a database, such as database **124**. In some embodiments, the security sub-system **114** may encrypt the XOR snapshot **112** when it is output to the external device **130**. In some embodiments, the security sub-system **114** determines that the external device **130** has Bluetooth® capabilities prior to outputting the XOR snapshot **112**. For example, the security sub-system **114** may determine that Bluetooth® is enabled on the external device **130** and/or the host device **120**.

(41) FIG. **6** is a flowchart illustrating a method **600** for storing parity information in the security sub-system memory **117**, according to some embodiments. It is understood that the process **600** may be stored in a memory, such as memory device **104**, and executed by a processor, such as processor **119**. However, it is contemplated that some or all of the process **600** may be performed on the memory device **104**, such as via the read/write circuitry **111**. Accordingly, while the process **600** is described in regards to the controller **106**, and in particular, the security sub-system **114**, and its associated components described above, it is contemplated that the process **600** may be at least partially performed by the memory device **104** or other suitable memory component. In some embodiments, the process **600** may be executed in conjunction with process **500**.

(42) The security sub-system **114** receives parity information from the storage sub-system **108** (at block **602**). For example, the parity information may be a XOR snapshot, such as XOR snapshot **112**. In some embodiments, the security sub-system **114** receives parity information from the storage sub-system **108** in response to the storage sub-system **108** receiving a memory transfer request. In some embodiments, the XOR snapshot **112** is received by the Bluetooth® controller **115** of the security sub-system **114**. For example, the memory **110** of the storage sub-system **108** may send the XOR snapshot to the Bluetooth® controller **115** and, in particular, a Bluetooth® transceiver of the Bluetooth® controller **115**.

(43) Once the security sub-system **114** receives the parity information from the storage sub-system **108**, the security sub-system **114** stores the parity information in a security sub-system memory, such as security sub-system memory **117** (at block **604**). For example, the security sub-system memory **117** may be a SRAM. Alternatively, or additionally, in some embodiments, the XOR snapshot **112** may still be stored in the memory device **104** at a predetermined rate that is lower than conventional rates, where the predetermined rate reduces some of the overhead experienced by the memory device **104** with respect to the conventional rates. In some embodiments, the memory device **104** may be NAND. In some embodiments, the XOR snapshot **112** is stored in the NAND in a non-blocking manner.

(44) FIG. 7 is a flowchart illustrating a method **700** for retrieving parity information, according to some embodiments. It is understood that the process **700** may be stored in a memory, such as memory device **104**, and executed by a processor, such as processor **119**. However, it is contemplated that some or all of the process **700** may be performed on the memory device **104**, such as via the read/write circuitry **111**. Accordingly, while the process **700** is described in regards to the controller **106**, and in particular, the security sub-system **114**, and its associated components described above, it is contemplated that the process **700** may be at least partially performed by the memory device **104** or other suitable memory component.

(45) The security sub-system **114** receives a parity information request from the storage sub-system **108** (at block **702**). In some embodiments, the parity information request is a request for the retrieval of the XOR snapshot **112**. In some embodiments, the security sub-system **114** receives the parity information request from the storage sub-system **108** in response to the storage sub-system initiating a parity information retrieval request. For example, the parity information retrieval request may be initiated when a read command results in a data disturbance and the parity information **112** from the storage sub-system **108** may be used to compute the data back (e.g., recover data by reducing or eliminating the data disturbance with respect to disturbed data). When the security sub-system **114** receives a parity information request from the storage sub-system **108**, the security sub-system **114** determines whether the parity information is available in the security sub-system memory **117** or from the external device **130** (at decision block **704**). In some embodiments, the security sub-system first checks the security sub-system memory **117** to determine if the parity information is stored therein. When the security sub-system **114** determines that the parity information is available from the external device ("External Device" at decision block **704**), the method **700** proceeds to block **706**. When the security sub-system **114** determines that the parity information is available in the security sub-system memory **117** ("Memory" at decision block **704**), the method proceeds to block **710**.

(46) At block **706**, the security sub-system **114** requests the parity information from the external device **130**. In some embodiments, a Bluetooth® transceiver of the Bluetooth® controller **115** may send a request to the wireless communication controller **402** of the external device **130** for the external device **130** to execute the application **122**. For example, the external device **130** may execute the application **122** which retrieves the XOR snapshot **112** from the database **124** and/or the memory **404**. The security sub-system **114** receives the parity information from the external device **130** (at block **708**). In some embodiments, the Bluetooth® transceiver of the Bluetooth® controller **115** receives the XOR snapshot **112** from the memory **404** of the external device **130** via the wireless communication controller **402** of the external device **130**. In response to receiving the parity information from the external device **130**, the security sub-system **114** outputs the parity information to the storage sub-system **108** (at block **712**). In some embodiments, the Bluetooth® transceiver of the Bluetooth® controller **115** outputs the XOR snapshot **112** to the memory **110** of the storage sub-system **108**.

(47) At block **710**, the security sub-system **114** retrieves the parity information from the security sub-system memory **117**. In some embodiments, the Bluetooth® transceiver of the Bluetooth® controller **115** retrieves the XOR snapshot **112** from the security sub-system memory **117**. In response to retrieving the parity information from the security sub-system memory **117**, the security sub-system **114** outputs the parity information to the storage sub-system **108** (at block **712**). In some embodiments, Bluetooth® transceiver of the Bluetooth® controller **115** outputs the XOR snapshot **112** to the memory **110** of the storage sub-system **108**. In some embodiments, the data storage device controller **106** performs parity operations with the retrieved parity information when the storage sub-system **108** receives the parity information. For example, parity operations include data recovery operations, data correction operations, or other suitable operations.

(48) FIG. 8 is a flowchart illustrating a method **800** of storing parity information in the external device **130** of FIG. 1, according to some embodiments. It is understood that the process **800** may be

stored in a memory, such as memory **404**, and executed by a processor, such as processor **400**. However, it is contemplated that some or all of the process **800** may be performed on the memory **404**, such as via read/write circuitry. Accordingly, while the process **700** is described in regards to the external device **130** and its associated components described above, it is contemplated that the process **800** may be at least partially performed by the memory **404** or other suitable memory component.

(49) The external device **130** receives parity information from a security sub-system, such as security sub-system **114** (at block **802**). In some embodiments, a wireless communication controller of the external device **130**, such as wireless communication controller **402**, communicates with the Bluetooth® controller **115** of the security sub-system **114** to receive an XOR snapshot **112**. For example, the wireless communication controller **402** may include a Bluetooth® controller including a Bluetooth® transceiver. In some embodiments, the parity information is encrypted when received from the security sub-system **114**. When the wireless communication controller **402** receives the parity information from the security sub-system **114**, the electronic processor **400** of the external device **130** stores the parity information in the memory **404** (at block **804**). In some embodiments, the memory **404** includes an application **122** that, when executed by the electronic processor **400**, stores the XOR snapshot **112** in the memory **404**. For example, the application **122**, when executed by the electronic processor **400**, may store the XOR snapshot **112** in a database **124** within the application **122**.

(50) FIG. **9** is a flowchart illustrating a method **900** of outputting parity information from the external device **130** of FIG. **1**, according to some embodiments. It is understood that the process **800** may be stored in a memory, such as memory **404**, and executed by a processor, such as processor **400**. However, it is contemplated that some or all of the process **800** may be performed on the memory **404**, such as via read/write circuitry. Accordingly, while the process **700** is described in regards to the external device **130** and its associated components described above, it is contemplated that the process **800** may be at least partially performed by the memory **404** or other suitable memory component.

(51) The external device **130** receives a parity information request from a security sub-system, such as security sub-system **114** (at block **902**). In some embodiments, a wireless communication controller, such as wireless communication controller **402**, communicates with the Bluetooth® transceiver of the Bluetooth® controller **115** of the security sub-system **114** to receive the parity information request. For example, the security sub-system **114** may wish to retrieve the XOR snapshot **112** from the external device **130**. In some embodiments, the external device **130** receives a first identifier (e.g., a universally unique identifier (“UUID”), token, or other suitable identifying information) that is associated with the parity information that the security sub-system **114** wishes to retrieve along with, or subsequent to the parity information request. For example, the first identifier may be specific to the XOR snapshot **112** that the security sub-system **114** is requesting from the external device **130**. In some embodiments, the first identifier may be generated, maintained, and provided by the security sub-system **114** along with the parity information when the parity information is received by the external device **130**. In other embodiments, the first identifier may be generated, maintained, and provided by the storage sub-system **108**.

(52) In some embodiments, the electronic processor **400** of the external device **130** stores the first identifier in the memory **404** along with the parity information. When the external device **130** receives the parity information request from the security sub-system **114**, the electronic processor **400** of the external device **130** determines that the parity information is available in the memory **404** of the external device **130** (at block **904**). In some embodiments, the electronic processor **400** executes the application **122** (stored within the memory **404**) to determine that the XOR snapshot **112** is within memory **404**. The electronic processor **400** may use the first identifier to locate the parity information in the memory **404** of the external device **130**. In some embodiments, the electronic processor **400** matches that first identifier to an identifier (i.e. the stored first identifier)

of the parity information in the memory **404** to determine that the requested parity information (i.e., the parity information requested by the parity information request) is in the memory **404**. In some embodiments, the external device **130** may store first parity information from a first data storage device and second parity information of a second data storage device that is distinct from the first parity information. In that case, the first identifier may be associated with the particular data storage device that output the parity information request.

(53) When the electronic processor **400** determines that the parity information is available in the memory **404**, the external device **130** outputs the parity information to the security sub-system **114** (at block **906**). In some embodiments, electronic processor **400** may execute the application **122** which outputs the XOR snapshot **112** via the wireless communication controller **402**. For example, the wireless communication controller **402** outputs the XOR snapshot **112** to the Bluetooth® transceiver of the Bluetooth® controller **115** of the security sub-system **114**. In some embodiments, the output parity information is encrypted by the external device **130** and decrypted when received by the security sub-system **114**.

(54) The Abstract is provided to allow the reader to quickly ascertain the nature of the technical disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. In addition, in the foregoing Detailed Description, it can be seen that various features are grouped together in various embodiments for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the claimed embodiments require more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter lies in less than all features of a single disclosed embodiment. Thus, the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separately claimed subject matter.

Claims

1. A data storage device, comprising: a memory; a data storage device controller coupled to the memory and including a storage sub-system and a security sub-system, the storage sub-system including parity information within a second memory; a first bus configured to facilitate a transfer of a first copy of the parity information between the data storage device controller and a dedicated memory of an external device in communication with the data storage device controller, wherein the dedicated memory is dedicated to storing parity information; and a second bus configured to receive host commands, send responses to the host commands, and facilitate a transfer of a second copy of the parity information between the data storage device controller and a host device in communication with the data storage device controller, the host device being separate and distinct from the external device, wherein the data storage device controller is configured to: initiate a parity information back-up request, send, from the second memory of the storage sub-system, the parity information to the security sub-system based on the initiation of the parity information back-up request, and control the security sub-system to transmit the first copy of the parity information to the external device.
2. The data storage device of claim 1, wherein the security sub-system wirelessly transmits the first copy of the parity information to the dedicated memory of the external device with a Bluetooth® transceiver, and wherein the first bus is coupled to the security sub-system.
3. The data storage device of claim 1, wherein the parity information back-up request is initiated by the data storage device controller in response to a power down of the data storage device.
4. The data storage device of claim 1, wherein the parity information back-up request is initiated by the data storage device controller in response to at least one of a set fixed time interval or a power down of the data storage device.
5. The data storage device of claim 4, wherein the set fixed time interval is in a range of 0.5 seconds to 30 seconds.

6. The data storage device of claim 1, wherein the storage sub-system sends the parity information to the security sub-system via an internal bus.
7. The data storage device of claim 1, wherein the security sub-system encrypts the first copy of the parity information prior to providing the first copy of the parity information to the dedicated memory of the external device with a Bluetooth® transceiver.
8. The data storage device of claim 1, wherein the data storage device controller is further configured to: provide, from the second memory of the storage sub-system, the parity information to a third memory of the security sub-system, and store the parity information in the third memory of the security sub-system.
9. The data storage device of claim 8, wherein the data storage device controller is further configured to: initiate a parity information retrieval request, request, with the storage sub-system, the parity information from the security sub-system, retrieve, with the security sub-system, the parity information from the third memory, output, with the security sub-system, the parity information to the storage sub-system, and perform, with the storage sub-system, a parity operation with the parity information that is output.
10. The data storage device of claim 1, wherein the data storage device controller transmits the first copy of the parity information from a third memory of the security sub-system to the dedicated memory of the external device.
11. A method of backing-up parity information, the method comprising: initiating, with an electronic processor, a parity information back-up request; controlling, with the electronic processor, a storage sub-system coupled to the electronic processor to send parity information from a memory of the storage sub-system to a security sub-system coupled to the electronic processor based on the initiation of the parity information back-up request; controlling, with the electronic processor, the security sub-system to transmit a first copy of the parity information to a dedicated memory of an external device via a first bus; and controlling, with the electronic processor, a host interface coupled to a second bus to transmit a second copy of the parity information to a host device that is separate and distinct from the external device, wherein the dedicated memory is dedicated to storing parity information, and wherein the second bus is configured to receive host commands, send responses to the host commands, and facilitate the transmission of the second copy of the parity information to the host.
12. The method of claim 11, further comprising: controlling the storage sub-system to provide the parity information from the memory of the storage sub-system to a memory of the security sub-system; and controlling the security sub-system to store the parity information in the memory of the security sub-system.
13. The method of claim 12, further comprising: initiating a parity information retrieval request; controlling the storage sub-system to request the parity information from the security sub-system based on the initiation of the parity information retrieval request; and determining whether the parity information is available in the memory of the security sub-system or whether the first copy of the parity information is available from the external device.
14. The method of claim 13, further comprising: controlling the security sub-system to retrieve, in response to determining that the parity information is available in the memory of the security sub-system, the parity information from the memory of the security sub-system; controlling the security sub-system to output the parity information to the storage sub-system; and controlling the storage sub-system to perform a parity operation with the parity information that is output.
15. The method of claim 13, further comprising: controlling the security sub-system to request, in response to determining that the first copy of the parity information is available from the external device, the first copy of the parity information from the external device; receiving the first copy of parity information from the external device; controlling the security sub-system to output the first copy of the parity information to the storage sub-system; and controlling the storage sub-system to perform a parity operation with the first copy of the parity information that is output.

16. A non-transitory computer readable medium including computer executable instructions that, when executed by an electronic processor, causes the electronic processor to perform a set of operations comprising: initiating a parity information back-up request; controlling a storage sub-system to send parity information from a memory of the storage sub-system to a security sub-system based on the initiation of the parity information back-up request; controlling the security sub-system to transmit a first copy of the parity information to a dedicated memory of an external device via a first bus; and controlling a host interface coupled to a second bus to transmit a second copy of the parity information to a host device that is separate and distinct from the external device, wherein the dedicated memory is dedicated to storing parity information, and wherein the second bus is configured to receive host commands, send responses to the host commands, and facilitate the transmission of the second copy of the parity information to the host.

17. The computer readable medium of claim 16, wherein the computer executable instructions cause the electronic processor to perform the set of operations further comprising: controlling the storage sub-system to provide the parity information from the memory of the storage sub-system to a memory of the security sub-system; and controlling the security sub-system to store the parity information in the memory of the security sub-system.

18. The computer readable medium of claim 17, wherein the computer executable instructions cause the electronic processor to perform the set of operations further comprising: initiating a parity information retrieval request; controlling the storage sub-system to request the parity information from the security sub-system based on the initiation of the parity information retrieval request; and controlling the security sub-system to determine whether the parity information is available in the memory of the security sub-system or whether the first copy of the parity information is available from the external device.

19. The computer readable medium of claim 18, wherein the computer executable instructions cause the electronic processor to perform the set of operations further comprising: controlling the security sub-system to retrieve, in response to determining that the parity information is available in the memory of the security sub-system, the parity information from the memory of the security sub-system; controlling the security sub-system to output the parity information to the storage sub-system; and controlling the storage sub-system to perform a parity operation with the parity information that is output.

20. The computer readable medium of claim 18, wherein the computer executable instructions cause the electronic processor to perform the set of operations further comprising: controlling the security sub-system to request, in response to determining that the first copy of the parity information is available from the external device, the first copy of the parity information from the external device; controlling the security sub-system to receive the first copy of the parity information from the external device; controlling the security sub-system to output the first copy of the parity information to the storage sub-system; and controlling the storage sub-system to perform a parity operation with the first copy of the parity information that is output.
